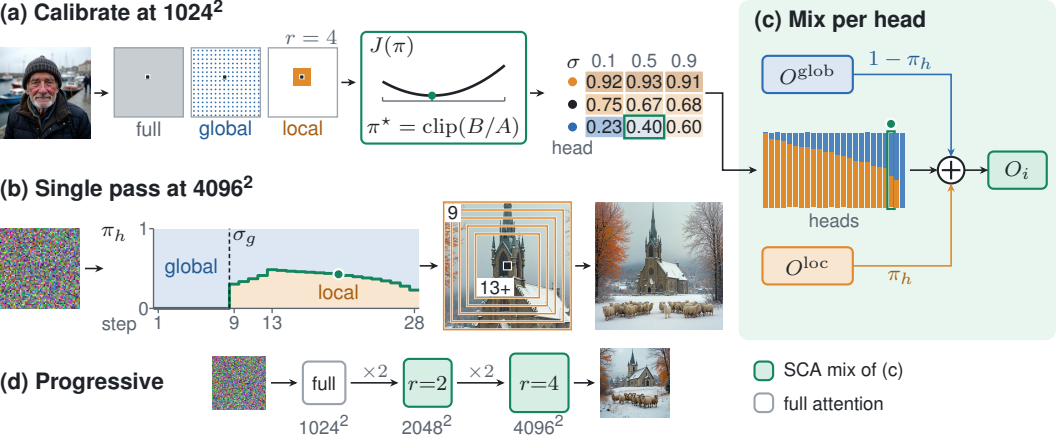

(a) Calibrate at 1024²
full
global
local
r = 4
J(π)
π* = clip(B/A)
σ 0.1 0.5 0.9
0.92 0.93 0.91
0.75 0.67 0.68
0.23 0.40 0.60
head
(c) Mix per head
O^glob
1 − π_h
heads
O_i
O^loc
π_h
(b) Single pass at 4096²
π_h
σ_g
global
local
step 1 9 13 28
9
13+
(d) Progressive
full
×2
r=2
×2
r=4
1024²
2048²
4096²
SCA mix of (c)
full attention